

A020 – The Stipulations: $\tilde{T} \cdot m = 1, T^4/Z_3, \xi = 4/30000$

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Contents

.1	Why the Stipulations Come First	1
.2	First Stipulation: Time-Mass Duality	1
.3	Second Stipulation: The Compact Geometry	1
.4	Third Stipulation: The Numerical Value	2
.5	What Follows from the Three Stipulations	3
.6	The Tolerance Standard at This Point	3
.7	Verification Scripts	3
.8	Open Questions	4
.9	Supersedes	4
.10	Use of AI Tools	4

.1 Why the Stipulations Come First

[STIPULATION] This document stands at the beginning of the content section because a theory is measured by the number and nature of its stipulations, not by the number of its results.

FFGFT claims to derive the constants of the Standard Model from a single dimensionless parameter. This claim is worth only as much as the list of what is additionally assumed is short and open. If the list is concealed, the one-parameter claim is worthless, because any amount of information can be hidden in an unstated assumption.

There are three stipulations. They are stated here in full, before any result is mentioned.

.2 First Stipulation: Time-Mass Duality

[STIPULATION] For the intrinsic time scale $\tilde{T}$ and the mass m of an object:

$$\tilde{T} \cdot m = 1.$$

This is not a definition of mass and not a definition of time. It is the statement that both quantities are two readings of the same thing: a small mass is a long intrinsic time, a large mass a short one. The equation is reciprocal, hence symmetric – neither side is the explanatory one.

In communications engineering terms, this is the Fourier duality between time and frequency, translated into physical language. That is precisely where it came from: the duality was the starting point of the theory, not its result.

[B] From this single equation it follows immediately that time is not an additional independent quantity. An object with mass carries its time scale with it; it needs no externally imposed clock. This is why the geometry manages with four and not five circles (A030), and why time sits inside the state space and not beside it (A060).

What is stipulated and not shown here: that the reciprocity holds exactly and not merely approximately; and that the constant on the right-hand side equals one. The one is a choice of units – any other value could be absorbed into the scales – but the exactness of the reciprocity is a substantive claim that is stipulated, not proved.

This stipulation is also why the theory cannot place itself as a norm above other frameworks: $\tilde{T} \cdot m = 1$ is a commitment, not a discovered law. One who does not share it makes no error but works in a different framework. The distinction between law, norm, and stipulation is developed in A200; it begins here.

.3 Second Stipulation: The Compact Geometry

[STIPULATION] The underlying space is a four-dimensional compact torus with a Z_3 identification, written T^4/Z_3 .

Three sub-decisions are embedded here, to be assessed separately.

Compact instead of infinite. This is the actual stipulation. It has an immediate consequence: a compact space has a discrete spectrum. It follows that masses appear as eigenvalues and need not be inserted as free parameters. This consequence is the entire reason for the stipulation – and it is simultaneously the point at which the stipulation can fail.

Torus instead of sphere. [B] This choice is not free: the torus has a non-trivial fundamental group, $\pi_1(T^4) \cong \mathbb{Z}^4$, the sphere does not. Winding numbers – the discrete, integer, coordinate-free quantities on which the entire information concept of the theory rests (A180) – do not exist on the sphere. Replacing the torus by a simply connected surface loses not a detail but the load-bearing element.

Four dimensions. The accounting here is uncomfortable, and should remain so. That there are *four* and not five follows from the first stipulation: because $\tilde{T} \cdot m = 1$ binds time to mass, no separate fifth circle for time is needed. In this sense the four is derived. But that the experienced structure is precisely $3 + 1$ – three spatial directions and one time – is **empirical input** and is not derived by the theory. Earlier formulations in the corpus were too strong at this point; the correction is recorded in the register as P34 and incorporated here.

The Z_3 identification. [STIPULATION] The tripling is stipulated. It delivers the three-element fibre on which the three generations appear as the spectrum of a cyclic operator (A110). **[B]** What can be shown: tripling is lattice-compatible. On T^4 , however, $n = 5$ is also lattice-compatible (e.g. the A_4 lattice); the admissible orders are $n \in \{1, 2, 3, 4, 5, 6, 8, 10, 12\}$. The 2D restriction $2 \cos(2\pi/n) \in \mathbb{Z}$ (with $n \in \{1, 2, 3, 4, 6\}$) does not single out $n = 3$ on T^4 ; 5 would be excluded only for a hypercubic metric. *(Corrected on 1 Oct. 2026: previously “tripling is crystallographically permitted, fivefold symmetry is not – the condition $2 \cos(2\pi/n) \in \mathbb{Z}$ permits $n \in \{1, 2, 3, 4, 6\}$ ” – this restriction holds only for lattices in 2 and 3 dimensions; the companion matrix of Φ_5 is integral with $\det = 1$ and $C^5 = I$, so $\mathbb{Z}^4$ has an automorphism of order 5; the orders in $GL(4, \mathbb{Z})$ are $\{1, 2, 3, 4, 5, 6, 8, 10, 12\}$, for a hypercubic T^4 (point group B_4) $\{1, 2, 3, 4, 6, 8\}$.)* This does not explain why three must be realised; but it narrows the alternatives.

[STIPULATION] The direction is to be noted: $n = 3$ is *fixed* not by symmetry but by the observed number of generations – the observation selects the symmetry, not the other way around. Z_3 is stipulated, motivated by the economy of the three-point configuration and consistent with the generation count. The reversal – “three generations are not free parameters but the three permitted Z_3 fixed points” – reads the observation as a consequence of the symmetry and is to be avoided. A stipulation does not become a derivation by the presence of reasons.

.4 Third Stipulation: The Numerical Value

[STIPULATION] The parameter is

$$\xi = \frac{4}{30000} = \frac{1}{3 \cdot 2^2 \cdot 5^4} \approx 1.333 \cdot 10^{-4}.$$

This stipulation is the most vulnerable of the three, and it decomposes into two parts with quite different status. The separation is important enough to make explicit – earlier presentations in the corpus treated both parts jointly as derived, which was not sustainable (register P33).

The form is structural. [B] The factors $4/3$, the 3, and the 2^2 have a geometric origin: they come from the dimension structure and the tripling. In this respect the shape of the expression is not chosen.

The order of magnitude is not. The factor 5^4 – the question of why ξ sits at 10^{-4} and not at 10^{-2} or 10^{-6} – is *not* derived forward. It is justified in two ways, and both are honestly to be named for what they are: intuitively via harmonic ratio structures, and empirically via the Higgs sector. Neither justification is a derivation from the geometry.

[STIPULATION] Therefore: the numerical value of ξ is an input. The theory does not claim to generate it; it claims that everything else follows from it.

A warning belongs at the same place, because it concerns the most common error of reasoning. Every dimensionless quantity can be written as a power of ξ if one is free to choose the exponent. An expression of the form $X = \xi^N$ is in itself *vacuous*. It becomes a statement only when the exponent is fixed independently – by the geometry, by a count, by a recursion depth. Where this is not possible in this series, it will be said.

.5 What Follows from the Three Stipulations

[B] The three stipulations together fix: a compact space with discrete spectrum, a three-element fibre, a scale. Nothing more. Everything further in this series is either derived from these three, or explicitly marked as a further stipulation.

The test is the count itself. If in the course of the series a fourth or fifth stipulation becomes necessary, it stands at its place with the marker **[STIPULATION]** – and it must then enter the balance of A230. A theory with three openly declared stipulations is better than one with a concealed one; one with seven openly declared would be worse than one with three. The accounting decides, not the rhetoric.

.6 The Tolerance Standard at This Point

This document deliberately contains not a single comparison number. The reason is methodological: stipulations have no measurement precision. It is meaningless to ask whether $\tilde{T} \cdot m = 1$ holds to six places – the equation is stipulated, not measured.

Stipulations are testable only indirectly, through their consequences, and there the standard from A010 applies: a consequence is considered supported when its residual stays within the determination tolerance and need not be closed by a free parameter. The first such numbers appear in A100.

.7 Verification Scripts

Two statements of this document are computational and therefore backed by scripts:

- the lattice compatibility of rotation orders: in 2D/3D via $2 \cos(2\pi/n) \in \mathbb{Z}$ with $n \in \{1, 2, 3, 4, 6\}$, on T^4 via $GL(4, \mathbb{Z})$ with $n \in \{1, 2, 3, 4, 5, 6, 8, 10, 12\}$, i.e. including fivefold symmetry (*Corrected on 1 Oct. 2026: previously "with the result $n \in \{1, 2, 3, 4, 6\}$ and the exclusion of fivefold symmetry" – the script checked only the 2D condition and has been extended to the 4D criterion.*): `python/A_Serie_Skripte/a020_kristallographie.py`
- the vacuousness of $X = \xi^N$ with a free exponent – shown on arbitrarily chosen target quantities for which a matching N can always be found: `python/A_Serie_Skripte/a020_xi_potenz_trivialitaet.py`

.8 Open Questions

First, the number three is not proved minimal. That exactly these three stipulations are needed is a claim about the economy of the construction. It has not been shown that none of them can be obtained from the others, nor that two would suffice.

Second, the tripling is permitted but not explained. The lattice compatibility on T^4 permits $\{1, 2, 3, 4, 5, 6, 8, 10, 12\}$, fivefold symmetry included. Why of the permitted values precisely three is realised, the geometry does not say. (*Corrected on 1 Oct. 2026: previously "The crystallographic restriction excludes fivefold symmetry and permits $\{1, 2, 3, 4, 6\}$ " – this restriction holds only in 2 and 3 dimensions, see above.*)

Third, the order of magnitude of ξ is justified but not yet derived within the A-series. The exponent -4 follows from $D_f = 3 - \xi$ and the four-dimensionality of the universe; $\kappa = 7$ emerges as the unique consistent solution of the e-p- μ system; the factor $4/3$ is stipulated or motivated, there is no Casimir confirmation. (*Removed on 1 Oct. 2026: previously "the factor $4/3$ is independently confirmed by the Casimir effect" – the standard Casimir energy $-\pi^2 \hbar c / (720 a^3)$ per area and the pressure $-\pi^2 \hbar c / (240 a^4)$ contain no factor $4/3$.*) This derivation is in A015. What *still* lacks: a derivation of the order of magnitude from the three stipulations themselves, without recourse to measured masses as input. Until then, the quantitative side carries a measured anchor.

Fourth, $3 + 1$ is an input. It is marked as such and will not be silently upgraded to a derivation in any later document of the series. Whether this is maintained, the audit script checks.

.9 Supersedes

This document subsumes: Doc. 003 (foundations), Doc. 008 (ξ and e), Doc. 009 (ξ -origin), Doc. 041 (parameter derivation), Doc. 042 (ξ parameter particles), Doc. 253 (ξ number space). Incorporated register entries: P33 (status of the magnitude), P34 (dimension structure), P35 (ξ^N triviality), R56 (Z_3 as stipulation; direction observation $\rightarrow$ symmetry, not the reverse).

.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, the theoretical stipulations, and all substantive decisions come from the author; the tools formulate, compute, check for contradiction, and create verification scripts. They make no substantive decisions. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.